

Paper Reading Notes

Building Services and Information

Building Science / Building Information
Screening and Literature Library

Paper ID: permanent library identifier (P001, P002, ...).

BibTeX key: bibliography identifier.

Citation number: assigned in citation order; independent of Paper ID.

目次

1	A. Building Environment and IEQ	3
	P001 — Spatial-temporal CO2 concentration prediction in education facilities with dynamic layouts using BIM and multi-modal graph neural networks	3
	P002 — Graph-based spatial-temporal prediction and feature interaction analysis of CO2 and occupant in large indoor space	3
	P005 — CO2 concentration prediction in office spaces using physics-informed neural network based on number of occupants and IoT sensor data	4
	P006 — Understanding the effect of outdoor pollution episodes and HVAC type on indoor air quality	4
	P014 — CO2- and PM2.5-Focused Optimal Ventilation Strategy Based on Predictive Control	4
2	B. HVAC and Building Services	6
	P015 — Deep reinforcement learning control for co-optimizing energy consumption, thermal comfort, and indoor air quality in an office building	6
3	C. Building Energy and ZEB	7
4	D. BMS, BAS, BACS and Building Automation	8
	P011 — Integrating the Brick Schema and Control Description Language for automation and controls in BIM	8
5	E. BIM and Semantic Information	9
	P009 — A systematic comparison and evaluation of building ontologies for deploying data-driven analytics in smart buildings	9
	P010 — BIM-to-BRICK: Using graph modeling for IoT/BMS and spatial semantic data interoperability within digital data models of buildings	9
6	F. Digital Twin	10
	P004 — How to measure and control indoor air quality based on intelligent digital twin platforms: A case study in China	10
	P007 — A BIM-enabled digital twin framework for real-time indoor environment monitoring and visualization by integrating autonomous robotics, LiDAR-based 3D mobile mapping, IoT sensing, and indoor positioning technologies	10
7	G. Sensors, IoT and Data Integration	11
	P013 — Smart metering data enhancement in sustainable buildings via knowledge graph-guided graph neural networks	11
8	H. Fault Detection, Diagnostics and Maintenance	12
9	I. AI and Machine Learning for Buildings	13

10	J. Retrieval, Query, LLM and Agent	14
	P012 — An ontology-aided, natural language-based approach for multi-constraint BIM model querying	14
11	K. Occupancy and Human-Building Interaction	15
12	L. Reviews, Frameworks and Standards	16
	P003 — Digital twin applications for indoor environmental quality in healthy building research: a systematic review of indicators, methods and computational paradigms	16
	P008 — Digital twin-enabled smart facility management: A bibliometric review	16
13	Uncategorized	17

1 A. Building Environment and IEQ

P001 — Spatial-temporal CO2 concentration prediction in education facilities with dynamic layouts using BIM and multi-modal graph neural networks

Authors: Hao Liu, Kam-Ming Mark Tam, Cong Huang, Chun Yin Lai, Eric Schuldenfrei

Major Category: Building Environment and IEQ

Secondary Category: AI and Machine Learning for Buildings, BIM and Semantic Information

Keywords: Spatial-temporal prediction, CO2 concentration, Education facility, Space reconfigurability, BIM, Graph neural networks, Multi-modal deep learning

■**主な主張** BIM 由来の時変レイアウトと Voronoi ベースの CO2 センサ隣接グラフを multi-modal deep learning で統合し, 2 時間先予測で MAE 11.25 ppm を達成し, 比較対象中で次点の TGCN の 15.10 ppm より約 25% 低い MAE を示した. [1]

■**新規性** 静的レイアウトを前提とする既往の GNN 型 CO2 予測に対し, BIM から時変の間仕切り構成をテンソル化し, センサグラフ文脈と空間構成文脈を二枝の MMDL で統合した点に新規性がある.

■**位置付け** 教育施設の IAQ 予測を主目的としつつ, BIM 幾何情報を予測モデルの入力文脈として利用するため, Building Environment and IEQ を主分類, AI and Machine Learning for Buildings および BIM and Semantic Information を副分類として位置付ける.

P002 — Graph-based spatial-temporal prediction and feature interaction analysis of CO2 and occupant in large indoor space

Authors: Cong Huang, Helen H.L. Kwok, Kwok Ho Poon, Zhaoji Wu, Fangli Hou, Jun Ma, Jack C.P. Cheng

Major Category: Building Environment and IEQ

Secondary Category: AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration

Keywords: Graph neural network, Spatial-temporal prediction, Spatial-temporal feature interactions, Indoor Air Quality, Indoor occupant, Dynamic graph learning, Occupant-in-loop

■**主な主張** 提案 AST-GCGRU は, 6 h および 12 h の multi-step prediction で graph-level R2 0.898 および 0.812 を示し, GCGRU の 0.860 および 0.699 を上回った. 12 h では occupant node R2 0.59, CO2 node R2 0.73 で, GCGRU の 0.32, 0.58 より高かった. [2]

■**新規性** 距離ベースの固定 graph ではなく, partial correlation で初期化した graph を ST-Att で学習中に更新し, time-lag nodes により 30 min CO2 sensor と 6 min occupant sensor の異なる時間解像度を保持したまま, CO2 と occupancy を同一 graph で同時予測・interaction analysis する構成を提示した.

■**位置付け** 大空間における IAQ と occupancy の同時予測・相互作用分析を主目的とするため, Building Environment and IEQ を主分類とし, AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration を副分類として位置付ける.

P005 — CO2 concentration prediction in office spaces using physics-informed neural network based on number of occupants and IoT sensor data

Authors: Jehyun Kim, Gihoon Kim, Jong-Il Bang, Anseop Choi, Minki Sung

Major Category: Building Environment and IEQ

Secondary Category: AI and Machine Learning for Buildings, Occupancy and Human-Building Interaction, Sensors, IoT and Data Integration

Keywords: CO2 prediction, Physics-informed neural network, Occupancy, HVAC operation, IoT, Indoor air quality

■**主な主張** occupancy と HVAC/IoT data に CO2 mass-balance 制約を加えた PINN で office CO2 を予測した. [3]

■**新規性** data-driven CO2 prediction へ mass-balance equation を physics-informed loss として導入した.

■**位置付け** CO2 prediction を中心とする IEQ × PINN 研究として位置付ける.

P006 — Understanding the effect of outdoor pollution episodes and HVAC type on indoor air quality

Authors: Tristalee Mangin, Zachary Barrett, Zachary Palmer, Dillon Tang, Sean Nielson, Darrah Sleeth, Kerry Kelly

Major Category: Building Environment and IEQ

Secondary Category: HVAC and Building Services, Sensors, IoT and Data Integration

Keywords: PM2.5, HVAC, Infiltration, Wildfire smoke, Indoor air quality, Low-cost sensors

■**主な主張** 18 か月, 21 台の PM2.5 sensor で outdoor pollution episode と HVAC type による室内 PM2.5 infiltration 差を評価した. [4]

■**新規性** inversion, dust, wildfire smoke を HVAC type 別に long-term 比較した.

■**位置付け** outdoor-to-indoor PM2.5 infiltration と HVAC 影響の実測研究として位置付ける.

P014 — CO2- and PM2.5-Focused Optimal Ventilation Strategy Based on Predictive Control

Authors: Young Jae Choi, Eun Ji Choi, Jae Yoon Byun, Hyeun Jun Moon, Min Ki Sung, Jin Woo Moon

Major Category: Building Environment and IEQ

Secondary Category: HVAC and Building Services, AI and Machine Learning for Buildings

Keywords: CO2, PM2.5, Ventilation, Predictive control, VAV, Indoor air quality, Co-simulation

■**主な主張** CO2/PM2.5 予測を用いる VAV ventilation control を提案し, co-simulation で両汚染物質の制御性能を評価した. [5]

■**新規性** CO2 と PM2.5 の予測 error に応じ control priority を動的に切替えた.

■位置付け multi-pollutant IAQ predictive ventilation control 研究として位置付ける.

2 B. HVAC and Building Services

P015 — Deep reinforcement learning control for co-optimizing energy consumption, thermal comfort, and indoor air quality in an office building

Authors: Fangzhou Guo, Sang woo Ham, Donghun Kim, Hyeun Jun Moon

Major Category: HVAC and Building Services

Secondary Category: AI and Machine Learning for Buildings, Building Environment and IEQ, Building Energy and ZEB

Keywords: Deep reinforcement learning, HVAC control, Energy, Thermal comfort, CO₂, PM_{2.5}, DDPG

■**主な主張** DDPG-based DRL で HVAC を制御し, energy, thermal comfort, CO₂, PM_{2.5} を同時最適化した. [6]

■**新規性** 複数 HVAC actuator と IEQ-energy trade-off を単一 DRL supervisory controller で扱った.

■**位置付け** HVAC supervisory control における DRL-based multi-objective optimization 研究として位置付ける.

3 C. Building Energy and ZEB

4 D. BMS, BAS, BACS and Building Automation

P011 — Integrating the Brick Schema and Control Description Language for automation and controls in BIM

Authors: Esben Visby Fjerbæk, Kevin Michael Smith, Christian Anker Hviid

Major Category: BMS, BAS, BACS and Building Automation

Secondary Category: BIM and Semantic Information, HVAC and Building Services

Keywords: BACS, Brick Schema, Control Description Language, BIM, HVAC controls, Interoperability

■**主な主張** Brick Schema と CDL を統合し, BIM 由来 HVAC equipment と control logic を machine-readable knowledge graph で接続した. [7]

■**新規性** BACS context と control sequence を Brick+CDL で一体表現した.

■**位置付け** BACS/HVAC control semantics と BIM interoperability 研究として位置付ける.

5 E. BIM and Semantic Information

P009 — A systematic comparison and evaluation of building ontologies for deploying data-driven analytics in smart buildings

Authors: Zhangcheng Qiang, Stuart Hands, Kerry Taylor, Subbu Sethuvenkatraman, Daniel Hugo, Pouya Ghiasnezhad Omran, Madhawa Perera, Armin Haller

Major Category: BIM and Semantic Information

Secondary Category: Reviews, Frameworks and Standards, Sensors, IoT and Data Integration

Keywords: Building ontology, Data interoperability, Brick Schema, RealEstateCore, Project Haystack, Digital Buildings, Ontology evaluation

■**主な主張** Brick, RealEstateCore, Project Haystack, Digital Buildings の 4 ontology を TBox/ABox 両面で比較し, alignment の必要性を示した. [8]

■**新規性** axiomatic quality と実建物 data による completeness/expressiveness を同一 framework で評価した.

■**位置付け** smart-building ontology interoperability の比較評価研究として位置付ける.

P010 — BIM-to-BRICK: Using graph modeling for IoT/BMS and spatial semantic data interoperability within digital data models of buildings

Authors: Filippo Vittori, Chuan Fu Tan, Anna Laura Pisello, Adrian Chong, Cristina Piselli, Clayton Miller

Major Category: BIM and Semantic Information

Secondary Category: BMS, BAS, BACS and Building Automation, Sensors, IoT and Data Integration, Digital Twin

Keywords: BIM, Brick Schema, BMS, IoT, Knowledge graph, Data interoperability, Digital Twin

■**主な主張** BIM-to-BRICK により BIM 情報を Brick knowledge graph へ自動変換し, BIM と BMS/IoT data の semantic interoperability を実装した. [9]

■**新規性** BIM-to-Brick mapping を graph modeling で自動化した.

■**位置付け** BIM-BMS/IoT 間 semantic interoperability 研究として位置付ける.

6 F. Digital Twin

P004 — How to measure and control indoor air quality based on intelligent digital twin platforms: A case study in China

Authors: Yuchong Qian, Jiawei Leng, Kai Zhou, Yuxuan Liu

Major Category: Digital Twin

Secondary Category: Building Environment and IEQ, BIM and Semantic Information, Sensors, IoT and Data Integration

Keywords: Digital Twin, Indoor air quality, BIM, IoT, Machine learning, Monitoring and control

■**主な主張** IoT, BIM, machine learning を統合した IAQ 用 digital twin platform を構築し, monitoring, prediction, visualization, control を実装した. [10]

■**新規性** IAQ の Sense-Integrate-Retrieve-Analyze-Feedback を単一 DT platform へ統合した.

■**位置付け** BIM-IoT-ML 統合による IAQ Digital Twin platform 研究として位置付ける.

P007 — A BIM-enabled digital twin framework for real-time indoor environment monitoring and visualization by integrating autonomous robotics, LiDAR-based 3D mobile mapping, IoT sensing, and indoor positioning technologies

Authors: Xi Hu, Rayan H. Assaad

Major Category: Digital Twin

Secondary Category: Building Environment and IEQ, BIM and Semantic Information, Sensors, IoT and Data Integration

Keywords: Digital Twin, BIM, IoT, Indoor environmental monitoring, Indoor positioning, Robotics, LiDAR

■**主な主張** BIM, IoT, robotics, LiDAR, UWB positioning を統合し, real-time indoor environment monitoring を DT 上で可視化した. [11]

■**新規性** mobile robotic sensing と indoor positioning を BIM-enabled DT へ統合した.

■**位置付け** BIM-enabled spatial Digital Twin による IEQ monitoring 研究として位置付ける.

7 G. Sensors, IoT and Data Integration

P013 — Smart metering data enhancement in sustainable buildings via knowledge graph-guided graph neural networks

Authors: Fang He, Jiaqi Fan, Yang Deng, Xiaoyang Zhang, Ka Tai Lau, Dan Wang

Major Category: Sensors, IoT and Data Integration

Secondary Category: AI and Machine Learning for Buildings, BIM and Semantic Information

Keywords: Smart metering, Building knowledge graph, Graph neural network, Data enhancement, Knowledge embedding, Brick Schema

■**主な主張** building knowledge graph と GNN を融合した BuildKnow により smart-meter data enhancement を行い, 3 実建物で評価した. [12]

■**新規性** knowledge graph 由来の meter dependence を GNN の学習 prior として利用した.

■**位置付け** smart-meter data quality と knowledge-guided GNN 研究として位置付ける.

8 H. Fault Detection, Diagnostics and Maintenance

9 I. AI and Machine Learning for Buildings

10 J. Retrieval, Query, LLM and Agent

P012 — An ontology-aided, natural language-based approach for multi-constraint BIM model querying

Authors: Mengtian Yin, Llewellyn Tang, Chris Webster, Shen Xu, Xiongyi Li, Huaquan Ying

Major Category: Retrieval, Query, LLM and Agent

Secondary Category: BIM and Semantic Information

Keywords: Natural language query, BIM retrieval, Ontology, Semantic parser, SPARQL, IFC, Multi-constraint query

■**主な主張** ontology-aided semantic parser で multi-constraint natural language query を IFC-compliant SPARQL へ変換し, 225 NLQs で 91.11% の overall accuracy を報告した. [13]

■**新規性** 複合 NL query を ontology を介して structured BIM query へ変換した.

■**位置付け** 自然言語による BIM retrieval/query 研究として位置付ける.

11 K. Occupancy and Human-Building Interaction

12 L. Reviews, Frameworks and Standards

P003 — Digital twin applications for indoor environmental quality in healthy building research: a systematic review of indicators, methods and computational paradigms

Authors: Alessandro D’Amico, Edoardo Currà, Philomena M. Bluysen

Major Category: Reviews, Frameworks and Standards

Secondary Category: Digital Twin, Building Environment and IEQ, BIM and Semantic Information

Keywords: Digital Twin, Indoor environmental quality, Healthy buildings, BIM, IoT, Systematic review

■**主な主張** 62 件の DT × IEQ 研究を systematic review し, building semantics と occupant-related data の統合不足を整理した. [14]

■**新規性** DT × IEQ を指標, 方法, computational paradigm の観点から横断整理した.

■**位置付け** DT × IEQ の研究ギャップを整理する systematic review として位置付ける.

P008 — Digital twin-enabled smart facility management: A bibliometric review

Authors: Obaidullah Hakimi, Hexu Liu, Osama Abudayyeh

Major Category: Reviews, Frameworks and Standards

Secondary Category: Digital Twin, BIM and Semantic Information, Fault Detection, Diagnostics and Maintenance

Keywords: Digital Twin, Facility management, BIM, Semantic interoperability, Predictive maintenance, Autonomous control

■**主な主張** 248 件を bibliometric review し, DT-enabled FM の主要潮流と semantic interoperability, predictive maintenance 等の将来課題を整理した. [15]

■**新規性** DT-enabled FM 研究の構造と future needs を bibliometric に体系化した.

■**位置付け** Digital Twin × facility management の bibliometric review として位置付ける.

13 Uncategorized

References

- [1] Hao Liu, Kam-Ming Mark Tam, Cong Huang, Chun Yin Lai, and Eric Schuldenfrei. Spatial-temporal co2 concentration prediction in education facilities with dynamic layouts using bim and multi-modal graph neural networks. *Building and Environment*, 297:114528, 2026.
- [2] Cong Huang, Helen H.L. Kwok, Kwok Ho Poon, Zhaoji Wu, Fangli Hou, Jun Ma, and Jack C.P. Cheng. Graph-based spatial-temporal prediction and feature interaction analysis of co2 and occupant in large indoor space. *Building and Environment*, 280:112963, 2025.
- [3] Jehyun Kim, Gihoon Kim, Jong-Il Bang, Anseop Choi, and Minki Sung. Co2 concentration prediction in office spaces using physics-informed neural network based on number of occupants and iot sensor data. *Building and Environment*, 288:114035, 2026.
- [4] Tristalee Mangin, Zachary Barrett, Zachary Palmer, Dillon Tang, Sean Nielson, Darrah Sleeth, and Kerry Kelly. Understanding the effect of outdoor pollution episodes and hvac type on indoor air quality. *Building and Environment*, 278:112978, 2025.
- [5] Young Jae Choi, Eun Ji Choi, Jae Yoon Byun, Hyeun Jun Moon, Min Ki Sung, and Jin Woo Moon. Co2- and pm2.5-focused optimal ventilation strategy based on predictive control. *Indoor Air*, 2025(1):6652442, 2025.
- [6] Fangzhou Guo, Sang woo Ham, Donghun Kim, and Hyeun Jun Moon. Deep reinforcement learning control for co-optimizing energy consumption, thermal comfort, and indoor air quality in an office building. *Applied Energy*, 377:124467, 2025.
- [7] Esben Visby Fjerbæk, Kevin Michael Smith, and Christian Anker Hviid. Integrating the brick schema and control description language for automation and controls in bim. *Automation in Construction*, 184:106849, 2026.
- [8] Zhangcheng Qiang, Stuart Hands, Kerry Taylor, Subbu Sethuvenkatraman, Daniel Hugo, Pouya Ghiasnezhad Omran, Madhawa Perera, and Armin Haller. A systematic comparison and evaluation of building ontologies for deploying data-driven analytics in smart buildings. *Energy and Buildings*, 292:113054, 2023.
- [9] Filippo Vittori, Chuan Fu Tan, Anna Laura Pisello, Adrian Chong, Cristina Piselli, and Clayton Miller. Bim-to-brick: Using graph modeling for iot/bms and spatial semantic data interoperability within digital data models of buildings. *Energy and Buildings*, 348:116368, 2025.
- [10] Yuchong Qian, Jiawei Leng, Kai Zhou, and Yuxuan Liu. How to measure and control indoor air quality based on intelligent digital twin platforms: A case study in china. *Building and Environment*, 253:111349, 2024.
- [11] Xi Hu and Rayan H. Assaad. A bim-enabled digital twin framework for real-time indoor environment monitoring and visualization by integrating autonomous robotics, lidar-based 3d mobile mapping, iot sensing, and indoor positioning technologies. *Journal of Building Engineering*, 86:108901, 2024.
- [12] Fang He, Jiaqi Fan, Yang Deng, Xiaoyang Zhang, Ka Tai Lau, and Dan Wang. Smart metering data enhancement in sustainable buildings via knowledge graph-guided graph neural networks.

Knowledge-Based Systems, 325:114016, 2025.

- [13] Mengtian Yin, Llewellyn Tang, Chris Webster, Shen Xu, Xiongyi Li, and Huaquan Ying. An ontology-aided, natural language-based approach for multi-constraint bim model querying. *Journal of Building Engineering*, 76:107066, 2023.
- [14] Alessandro D’Amico, Edoardo Currà, and Philomena M. Bluysen. Digital twin applications for indoor environmental quality in healthy building research: a systematic review of indicators, methods and computational paradigms. *Building and Environment*, 304:115014, 2026.
- [15] Obaidullah Hakimi, Hexu Liu, and Osama Abudayyeh. Digital twin-enabled smart facility management: A bibliometric review. *Frontiers of Engineering Management*, 11(1):32–49, 2024.